Karthik Sengupta

sengupta.k@northeastern.edu | Boston, MA | linkedin.com/in/karthik-sengupta | github.com/kartsen03 | karthik-sengupta.vercel.app

Education

Northeastern University, Khoury College of Computer Sciences — Boston, MA
Master of Science in Computer Science | ***GPA: 4.0/4.0*** | *Expected Graduation: December 2027* — *Sep 2025 – Dec 2027*

- **Coursework:** Algorithms, Program Design Paradigms, Building Scalable Distributed Systems, Database Management Systems, Pattern Recognition & Computer Vision, Essentials of Data Science

Vellore Institute of Technology — Vellore, India
Bachelor of Technology in Computer Science and Engineering | *GPA: 3.55/4.0* — *Sep 2021 – May 2025*

- **Coursework:** Data Structures & Algorithms, Operating Systems, Computer Networks, Artificial Intelligence, Cloud Computing

Skills

Languages: C++ (C++17), Python, Java, Go, SQL, R
GPU & ML Systems: CUDA, NVRTC, cuBLAS, tensor cores (WMMA), Nsight Compute, Nsight Systems, PyTorch (incl. torch.compile), TensorFlow, Keras, scikit-learn
Vision & Data: OpenCV, MediaPipe, image quality assessment (PSNR, SSIM), time-series forecasting, anomaly detection, MySQL, PostgreSQL, SQLite, Power BI
Testing & DevOps: GoogleTest, pytest, JUnit, requirements traceability, GitHub Actions (CI/CD), Docker, Linux, CMake, Bash
AI-Assisted Development: Claude Code, with generated code verified by automated tests and CI

Professional Experience

Graduate Teaching Assistant — Program Design Paradigms (CS5010) — Sep 2026 – Present
Khoury College of Computer Sciences, Northeastern University — *Boston, MA*

- Review student code for design quality and test rigor (JUnit 5, JaCoCo, PIT mutation testing) in a graduate Java course on object-oriented design, and debug build and toolchain failures across Gradle, IntelliJ, and the JDK.

Research Intern — Jun 2024 – Jul 2024
National Center for Assistive Health Technology, IIT Delhi — *New Delhi, India*

- Designed, built, and deployed an AI application (OpenCV + LLM APIs) end to end for 200+ users across 8 sites, reaching 82% accuracy validated by domain specialists.
- Engineered Python pipelines and Flask REST APIs supporting concurrent multi-site access with production-grade error handling, cutting manual effort by 65%.

Software Development Engineer Intern — Aug 2023 – Oct 2023
LTIMindtree — *Chennai, India*

- Owned the employee onboarding and offboarding database end to end, from design through maintenance, and built the Power BI observability dashboards on top of it, replacing 15+ hours/month of manual reporting.
- Contributed SQL ingestion and automation improvements within the team's existing microservices platform, part of an effort that automated 500+ lifecycle events and cut onboarding time by 35%.

Projects

minicompiler: CUDA ML Graph Compiler | *C++17, CUDA, NVRTC, cuBLAS, Nsight* — GitHub

- Built a C++17 compiler for ML compute graphs (SSA DAG IR, dead-node elimination, constant folding, elementwise fusion, lifetime-based buffer planning) with Eigen CPU and CUDA backends, cutting an MLP block from 22 compute nodes to 4.
- Generated fused CUDA kernels with NVRTC, 10.1–10.5× faster than per-op kernels on 16–256 MB inputs on an RTX 3060 Laptop GPU, and wrote matmul kernels reaching 93% of cuBLAS at 2048^3, making a B=128 MLP block 1.86× faster than idiomatic PyTorch and 2.04× faster than torch.compile.
- Added FP16, BF16, and TF32 tensor-core kernels at 63–110% of cuBLAS, verified by 103 GoogleTest cases against a double-precision reference, with CI across GCC, Clang, ASan/UBSan, and two CUDA versions.

Validated Study Tracker | *Python, Flask, SQLAlchemy, SQLite, pytest* — GitHub

- Built a clinical study records app with an append-only audit trail enforced by database triggers, tracing 15 requirements to 219 tests at 100% line and branch coverage in a CI-generated matrix that fails the build on any untested requirement.

Clinical Trial Feasibility Explorer | *R, SQL, SQLite, GitHub Actions* — Live App | GitHub

- Built a validated data pipeline and interactive dashboard over 88,960 ClinicalTrials.gov trials, showing low accrual drove 32.4% of 11,855 terminated or withdrawn trials, deployed through tested CI/CD.

Battery RUL Prognostics | *Python, PyTorch, scikit-learn, statsmodels* — GitHub

- Built forecasting, remaining-useful-life, and fault-detection models over 770k sensor samples; found that cycles labeled as faults were a second test protocol and rebuilt the evaluation on injected faults (0.94 ROC-AUC).

Leadership, Awards & Publications

President, Khoury Masters Student Council (Apr 2026 – Present): Lead governance for 150+ MS students.
Global Student Mentor, Office of Global Services (Aug 2026 – Present): Mentor a cohort of incoming international students.
4th Place, GDG Cloud Boston Hackathon: Built Git-to-Doc, an LLM developer tool, with a 5-person team in 5 hours.
Publication: "RetailGPT: A Fine-Tuned LLM Architecture for Customer Experience and Sales Optimization," *ICSSAS 2024*.